

Lecture 7

Inner Product Spaces

The inner product is taken in the physics convention: conjugate-linear in the first slot, linear in the second, so $\langle \phi | \psi \rangle = \langle \phi, \psi \rangle$ is linear in the ket. Tags mark the flavour; a ★ marks a harder one.

Inner products and norms

Exercise 7.1 (computational) For $u = (1, -i, 2)$ and $v = (i, 1, 1 - i)$ in $\mathbb{C}^3$ with the standard inner product, compute $\langle u, v \rangle$, $\langle v, u \rangle$, $\|u\|$, and $\|v\|$.

Exercise 7.2 (proof) Prove that $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$ (the parallelogram law) in any inner product space.

Exercise 7.3 (computational) On $C[0, 1]$ with $\langle f, g \rangle = \int_0^1 \overline{f} g \, dt$, compute $\langle 1, t \rangle$, $\|1\|$, $\|t\|$, and verify the Cauchy–Schwarz inequality for these two functions.

Exercise 7.4 (computational) For $u = (1, 2, 2)$ and $v = (2, 2, 1)$ in $\mathbb{R}^3$ with the standard inner product, compute $\langle u, v \rangle$, $\|u\|$, $\|v\|$, and the cosine of the angle between them.

Exercise 7.5 (computational) For $u = (2, i)$ and $v = (1, 1 + i)$ in $\mathbb{C}^2$ with the standard inner product, compute $\langle u, v \rangle$, $\langle v, u \rangle$, $\|u\|$, and $\|v\|$.

Exercise 7.6 (computational) On $M_2(\mathbb{R})$ with the Frobenius inner product $\langle A, B \rangle = \text{tr}(A^\dagger B)$, take $A = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$. Compute $\langle A, B \rangle$, $\|A\|$, and $\|B\|$.

Exercise 7.7 (computational) On $C[0, 1]$ with $\langle f, g \rangle = \int_0^1 \overline{f} g \, dt$, let $f(t) = 1 + t$ and $g(t) = 1 - t$. Compute $\langle f, g \rangle$, $\|f\|$, and $\|g\|$.

Exercise 7.8 (proof) In a *real* inner product space, prove that if $\|u\| = \|v\|$ then $u + v$ is orthogonal to $u - v$ (the diagonals of a rhombus are perpendicular).

Dirac notation

Exercise 7.9 (computational) In an orthonormal basis $|0\rangle, |1\rangle$, let $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$. Find the probability of each outcome in a measurement in the $\{|0\rangle, |1\rangle\}$ basis, and the expectation value of $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$.

Exercise 7.10 (computational) In an orthonormal basis $|0\rangle, |1\rangle$, let $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$. Find the probability of each outcome in the $\{|0\rangle, |1\rangle\}$ basis and the expectation value of $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$.

Exercise 7.11 (computational) In an orthonormal basis $|0\rangle, |1\rangle$, let $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Find the probabilities of $|0\rangle, |1\rangle$ in the $\{|0\rangle, |1\rangle\}$ basis, the amplitude $\langle + | \psi \rangle$, and the probability $|\langle + | \psi \rangle|^2$.

Exercise 7.12 (computational) With $|0\rangle = (1, 0)$ and $|1\rangle = (0, 1)$, write the matrix of the outer product $|\psi\rangle\langle\psi|$ in this basis for $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

Exercise 7.13 (proof) Let $\{|n\rangle\}$ be an orthonormal basis and $|\psi\rangle = \sum_n c_n |n\rangle$. Prove that $\langle m | \psi \rangle = c_m$ and that $\|\psi\|^2 = \sum_n |c_n|^2$.

Exercise 7.14 (proof) Show that the bra associated to $a|\phi\rangle + b|\chi\rangle$ is $\bar{a}\langle\phi| + \bar{b}\langle\chi|$.

Orthogonality and Cauchy–Schwarz

Exercise 7.15 (computational) Find all $a \in \mathbb{C}$ for which $(1, a)$ is orthogonal to $(1, i)$ in $\mathbb{C}^2$.

Exercise 7.16 (computational) Find the value of $a \in \mathbb{R}$ for which $(1, a, 2)$ is orthogonal to $(3, -1, 1)$ in $\mathbb{R}^3$ with the standard inner product.

Exercise 7.17 (computational) In $\mathbb{R}^2$ decompose $u = (4, 2)$ as a multiple of $v = (1, 1)$ plus a vector orthogonal to v , using $\frac{\langle v, u \rangle}{\|v\|^2} v$ for the projection.

Exercise 7.18 (computational) For $u = (1, 1)$ and $v = (1, i)$ in $\mathbb{C}^2$, compute $|\langle u, v \rangle|$, $\|u\|$, and $\|v\|$, and verify the Cauchy–Schwarz inequality.

Exercise 7.19 (computational) In $\mathbb{R}^4$ find the cosine of the angle between $u = (1, 1, 1, 1)$ and $v = (1, 1, 1, -1)$, and hence the angle.

Exercise 7.20 (computational) Compute the Gram matrix $G_{jk} = \langle v_j, v_k \rangle$ of $v_1 = (1, 1, 0)$ and $v_2 = (0, 1, 1)$ in $\mathbb{R}^3$, and use it to decide whether they are linearly independent.

Exercise 7.21 (proof) Prove that if $u \perp v$ then $\|u + v\| \geq \|u\|$ and $\|u + v\| \geq \|v\|$.

Exercise 7.22 (proof) Use the Cauchy–Schwarz inequality to prove that for $a_1, \dots, a_n \in \mathbb{C}$,

$$\left| \sum_{k=1}^n a_k \right|^2 \leq n \sum_{k=1}^n |a_k|^2.$$

Exercise 7.23 (★) Prove that equality holds in the triangle inequality $\|u + v\| = \|u\| + \|v\|$ if and only if one of u, v is a nonnegative real multiple of the other.

Exercise 7.24 (★) Let $v_1, \dots, v_m$ be nonzero pairwise-orthogonal vectors. Prove they are linearly independent.

Exercise 7.25 (proof) For a subspace $U \subseteq V$ of an inner product space, prove that $U \cap U^\perp = \{0\}$.